

OpenClaw on Azure

Setup Guide for Docker Deployment

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Overview

This guide walks through provisioning an Azure VM, installing Docker, and deploying OpenClaw in a privileged Ubuntu container — giving OpenClaw full access to the VM while keeping your main Windows machine completely isolated.

This setup is recommended for developers who want to learn OpenClaw quickly without artificial permission restrictions, while keeping the host machine safe.

Recommended VM Specification

- Size: Standard_B2s — 2 vCPU, 4 GB RAM (~£15/month)
- Image: Ubuntu 24.04 LTS
- Disk: 30 GB Premium SSD
- Auth: SSH public key
- Region: UK South (or your nearest Azure region)

1 Provision the VM in Azure

Create a Resource Group and VM

Run the following commands using the Azure CLI. If you prefer the Portal, match these settings in the Create VM wizard.

```
az group create --name rg-openclaw --location uksouth

az vm create \
  --resource-group rg-openclaw \
  --name vm-openclaw \
  --image Ubuntu2404 \
  --size Standard_B2s \
  --admin-username azureuser \
  --generate-ssh-keys \
  --public-ip-sku Standard
```

Lock Down the Network Security Group

Allow inbound SSH only. OpenClaw's gateway is accessed via SSH tunnel, so no additional inbound ports are needed.

```
az network nsg rule create \  
  --resource-group rg-openclaw \  
  --nsg-name vm-openclawNSG \  
  --name AllowSSH \  
  --priority 1000 \  
  --protocol Tcp \  
  --destination-port-ranges 22 \  
  --access Allow
```

2 Install Docker on the VM

SSH into the VM, then install Docker using the official convenience script:

```
ssh azureuser@<your-vm-public-ip>  
  
# Install Docker  
curl -fsSL https://get.docker.com | sudo bash  
  
# Add your user to the docker group  
sudo usermod -aG docker azureuser  
  
# Apply group change without logging out  
newgrp docker  
  
# Verify  
docker --version  
docker compose version
```

3 Create the OpenClaw Docker Setup

Create a working directory and add the three files below:

```
mkdir ~/openclaw && cd ~/openclaw
```

Dockerfile

```
FROM ubuntu:24.04  
  
ENV DEBIAN_FRONTEND=noninteractive  
  
RUN apt-get update && apt-get install -y \  
  curl ca-certificates bash git \  
  && curl -fsSL https://deb.nodesource.com/setup_22.x | bash - \  
  && apt-get install -y nodejs \  
  && rm -rf /var/lib/apt/lists/*  
  
RUN npm install -g openclaw  
  
ENV OPENCLAW_STATE_DIR=/data/state  
ENV OPENCLAW_CONFIG_PATH=/data/config.json  
  
WORKDIR /data  
EXPOSE 18789  
  
CMD ["openclaw", "gateway", "--port", "18789"]
```

docker-compose.yml

```
services:
  openclaw:
    build: .
    container_name: openclaw
    restart: unless-stopped

    ports:
      - "127.0.0.1:18789:18789"    # Localhost only – use SSH tunnel

    environment:
      - ANTHROPIC_API_KEY=${ANTHROPIC_API_KEY}

    volumes:
      - openclaw_data:/data        # Persistent state
      - /:/host                    # Full VM filesystem access
      - /var/run/docker.sock:/var/run/docker.sock

    privileged: true
    network_mode: host

volumes:
  openclaw_data:
```

.env

```
ANTHROPIC_API_KEY=your_api_key_here
```

4 Build and Start OpenClaw

```
# Build the image and start the container
docker compose up -d --build

# Follow the logs
docker compose logs -f
```

5 Run the Onboarding Wizard

This configures your LLM provider and connects a chat channel (Telegram, Discord, WhatsApp, etc.):

```
docker exec -it openclaw openclaw onboard
```

For Telegram: the wizard will ask you to create a bot via @BotFather and paste the token in. This takes about 2 minutes.

6 Access the Control UI from Windows

Port 18789 is bound to localhost only on the VM. Use an SSH tunnel to access the Control UI securely from your Windows machine:

```
ssh -L 18789:localhost:18789 azureuser@<your-vm-public-ip> -N
```

Then open <http://localhost:18789> in your browser. If you use VS Code with the Remote SSH extension, it will offer to forward the port automatically.

Useful Commands

Container Management

```
# Stop OpenClaw
docker compose down

# Full clean rebuild (wipes state)
docker compose down -v && docker compose up -d --build

# Shell into the running container
docker exec -it openclaw bash

# Check gateway status
docker exec -it openclaw openclaw gateway status
```

Azure VM Management

```
# Stop the VM (saves compute cost)
az vm deallocate --resource-group rg-openclaw --name vm-openclaw

# Start it again
az vm start --resource-group rg-openclaw --name vm-openclaw

# Get current public IP (changes on restart unless using static IP)
az vm show -d --resource-group rg-openclaw --name vm-openclaw --query publicIps -o tsv
```

Security Considerations

This setup uses `privileged: true` and mounts the Docker socket, which means OpenClaw has full control over the VM. This is intentional for a learning environment on a dedicated single-purpose VM. Keep the following in mind:

- Do not install untrusted community skills from ClawHub without reviewing their source.
- Keep port 18789 closed on the NSG — always use the SSH tunnel.
- Your Azure VM is the security boundary, not the Docker container.
- Consider enabling Azure Defender for Servers if you store sensitive data on the VM.
- If you assign a static public IP in Azure, note that cost applies separately.